

The global, country-wise potential of green hydrogen in renewable-dominated electricity systems

Leon Schumm^{a,b,*}, Anton Achhammer^a, Michael Sterner^a

^a Institute for Energy Networks and Energy Storage (FENES), Faculty of Electrical and Information Technology, Ostbayerische Technische Hochschule Regensburg, Regensburg, Germany

^b Department of Digital Transformation in Energy Systems, Institute of Energy Technology, Fakultät III, Technische Universität Berlin, Germany

ARTICLE INFO

Keywords:

Green hydrogen
Renewable energy
Power-to-X
Electrolysis
Policy design
Energy transition
EU Delegated Act
PyPSA-Earth

ABSTRACT

The transition to clean fuels is essential for climate mitigation, with green hydrogen playing a key role. This study quantifies global, country-level export potentials under renewable electricity share requirements (as in the EU Delegated Act), considering ramp-up rates and domestic demand. The derived results for highly renewable countries are validated by comparing export quantities using the sector-coupled energy model PyPSA-Earth. These results are made accessible via a public dashboard. Findings show that the renewable share requirement significantly impacts hydrogen and Power-to-X export eligibility. A 90% minimum share allows 24 TWh of exports in 2025, mainly from South America and Europe, while lowering the threshold to 70% increases this 7.5-fold to 181 TWh. However, concerns about the emission intensity of lower thresholds persist. Export volumes rise further in the medium term (2030–2035), when renewable expansion outpaces demand growth. Results underscore the role of regulatory design in unlocking near-term hydrogen exports and supporting long-term decarbonization goals.

NOMENCLATURE

Symbols

$e_{t,c}$	Maximum expansion rate of technology t in country c
g_t	Growth rate of technology t
$d_{takeoff,t}$	Demand at which technology t surpasses 1% of total electricity supply
$e_{re,add,c}$	Additional renewable electricity generation in country c
a_{target}	Target year
$a_{current}$	Current year
TD	Threshold difference
s_{re}	Renewable electricity share
T	Renewable electricity share threshold (regulation)
e_{dom}	Total domestic electricity demand
$e_{exp,c}$	Available electricity for hydrogen exports in country c
$e_{dom,add,c}$	Additional domestic electricity demand in country c
$\eta_{electrolysis}$	Efficiency of electrolysis process
$h_{exp,c}$	Hydrogen export quantity of country c
Subscripts/Superscripts	
t	Technology
c	Country

(continued on next column)

(continued)

re	Renewable electricity
dom	Domestic electricity demand
exp	Electricity for export
add	Additional
takeoff	Takeoff year for technology
Abbreviations	
TD	Threshold difference
RE	Renewable Electricity
DA	European Commission's Delegated Act

1. Introduction

The global transition to clean fuels is a critical component of climate change mitigation strategies. While direct electrification is the most efficient pathway for many end uses; sectors such as steel, chemicals, and long-distance transport require energy-dense carriers or chemical feedstocks, making green hydrogen and its Power-to-X (PtX) products a critical component for decarbonizing hard-to-abate sectors [1–6].

This article is part of a special issue entitled: SDEWES-2025 (Kovač) published in International Journal of Hydrogen Energy.

* Corresponding author. Institute for Energy Networks and Energy Storage (FENES), Faculty of Electrical and Information Technology, Ostbayerische Technische Hochschule Regensburg, Regensburg, Germany.

E-mail addresses: leon1.schumm@oth-regensburg.de (L. Schumm), anton.achhammer@oth-regensburg.de (A. Achhammer), michael.sterner@oth-regensburg.de (M. Sterner).

<https://doi.org/10.1016/j.ijhydene.2026.156641>

Received 29 January 2026; Received in revised form 18 June 2026; Accepted 15 July 2026

0360-3199/© 2026 The Author(s). Published by Elsevier Ltd on behalf of Hydrogen Energy Publications LLC. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Key technologies, which are necessary for the transition to green hydrogen and PtX products (alkaline electrolysis, proton exchange membrane electrolysis, Haber-Bosch synthesis) have demonstrated high levels of technical readiness and efficiency [7–10]. However, despite ambitious targets, the global scaling-up of hydrogen production has not met expectations [11]. A key barrier is the slow deployment of electrolyzers [11], which require large volumes of low-carbon electricity to operate sustainably. While blue hydrogen — produced via steam methane reforming with carbon capture — offers an alternative for rapid productions scale-ups [12], concerns remain about long-term fossil fuel lock-in and residual greenhouse gas emissions [13]. In this context, regulatory frameworks play a crucial role in shaping the pace and direction of hydrogen and PtX development. For instance, the European Union, which is set to become a key importer of hydrogen and PtX products [14], has defined green hydrogen and PtX products in its European Commission's Delegated Act (DA), which mandates requirements for additionality, temporal, and geographical correlation unless the electricity system reaches a 90% renewable electricity (RE) share [15]. The requirements for additionality and correlation prove to be a major and costly barrier for hydrogen ramp-up [16], while the renewable share regulation potentially offers a pragmatic approach to support electrolyser deployment in highly decarbonized electricity systems. This has been seen in Denmark's DK1 bidding zone, where actors are currently focused on reaching the 90% renewable share while largely setting aside compliance with the other regulatory criteria for the time being [17,18].

This example illustrates the practical relevance of minimum RE share requirements as a near-term regulatory pathway, yet a globally consistent, quantitative assessment of the resulting hydrogen and PtX export potentials remains lacking in the literature.

Previous studies have analysed green hydrogen and PtX potentials under technical or resource constraints, often on a regional or national scale [19–27]. These studies have largely concentrated on assessing the techno-economic feasibility of hydrogen production, frequently neglecting the dynamic interplay between regulatory requirements and the development of national electricity systems. Studies focusing on the regulatory aspects as the DA [28–31], lack attention towards the 90% RE threshold. While efforts are made to model the 90% RE threshold in defossilised grids [32], a recent study [33] acknowledges the 90% threshold but explicitly excludes it from its scope, arguing that no European country currently meets this criterion, aside from individual bidding zones in Norway [34]. However, this perspective overlooks, that the regulation potentially provides a promising strategy for hydrogen-exporting countries which are already above the 90% threshold today or on the trajectory to reach that threshold [25].

While some efforts have been made to integrate policy considerations and electricity mix dynamics, there remains a notable gap in globally consistent, quantitative analyses that assess hydrogen and PtX export potential under minimum RE share requirements.

This study addresses this gap through three specific contributions.

First, this study contributes to the debate on hydrogen regulation by a globally consistent assessment of green hydrogen and PtX export potentials under differing RE share thresholds of 90% and 70%. The 90% threshold is directly derived from the EU Delegated Act, which exempts electricity systems exceeding this share from additionality, temporal, and geographical correlation requirements [15]. The 70% threshold reflects an industry proposal from Hydrogen Europe [35] and other stakeholders [34], who argue that the 90% cut-off is too stringent and propose a more relaxed rule of 70% to encourage electrolyser deployment in countries with rapidly growing renewable shares [34]. By examining both thresholds alongside intermediate values, this study systematically quantifies how regulatory ambition translates into exportable hydrogen and PtX quantities at the global level.

Second, by quantifying RE ramp-up rates and electricity system sizes consistently across all countries, we determine how more relaxed or stricter versions of the EU regulatory framework affect electrolysis-based hydrogen and PtX export potential at a global scale, an analysis

not previously available in the literature.

Third, using the open-source, sector-coupled energy model PyPSA-Earth, we validate these global results against a detailed capacity expansion and dispatch model of Ethiopia, a country that already fulfils the EU DA requirement today, among other countries assessed globally (s. Sec. RESULTS).

This combination of global consistency and country-level model validation distinguishes our approach from prior export-potential studies, which offer either highly aggregated global estimates or detailed but country-specific assessments without a unifying regulatory framework.

Fig. 1 displays the share of electricity generated from low-carbon sources by country. Countries outlined in red have a share of 90% or more, making them eligible to export green hydrogen under a 90% minimum RE share requirement (unless the low-carbon share is reached based on nuclear power, which is at least not included in the EU's DA). A total of 19 countries meet this threshold, with the majority (10) located on the African continent. While these countries tend to have relatively low electricity demand [38], their high renewable shares position them well for future green hydrogen exports. In contrast, several of the world's largest electricity systems (including China, the United States, India, Russia, and Japan) currently remain below 35% RE. As discussed in the Section METHODS, countries combining high renewable shares with large electricity systems (such as Norway, Brazil, and Canada) are particularly well suited for substantial green hydrogen export volumes under minimum RE share requirements.

However, the number of countries fulfilling those prerequisites is small, raising the question whether a minimum share regulation may substantially contribute to global hydrogen and PtX quantities and unlock potentials in the short-term. To address this question, this study outlines the key factors on determining the global, country-wise hydrogen and PtX export potential under minimum RE share requirements and the methodological framework used for the sector-coupled energy model demonstrated in Ethiopia and further countries with high RE shares (s. Sec. METHODS). Furthermore, the Section RESULTS investigates how a relaxed regulation or future RE expansions leverage hydrogen and PtX exports while paying special attention to the domestic electricity demand and carbon intensity, further elaborated in the Section DISCUSSION. Finally, the Section CONCLUSION summarizes the main findings.

In this study, the term “countries” is used in accordance with the ISO 3166-1 alpha-2 standard. This includes not only internationally recognized sovereign states but also dependent territories and special areas of geographical interest, as listed in the ISO classification [39].

2. Methods

To quantify the global, country-wise hydrogen and PtX export potential, Section METHODS presents global datasets on electricity demand, applies methods of RE expansion and determines the global export quantity. Based on determining PtX export potentials on a national level presented in Ref. [25], we expand the methods to apply on a global level. Second, the energy model PyPSA-Earth, used for validating the results by investigating Ethiopia and further countries, is presented.

2.1. Electricity demand, renewable electricity expansion, hydrogen and PtX export quantity

To quantify the hydrogen export quantity, this study applies a global dataset on electricity demand, adapts a method to quantify the RE expansion and calculates the hydrogen export quantity taking the minimum RE share requirement into account.

Domestic electricity demand. In many countries globally, domestic electricity demand is projected to increase substantially due to ongoing electrification trends (such as in the transport and heating sectors) alongside population and economic growth. Additional electricity loads

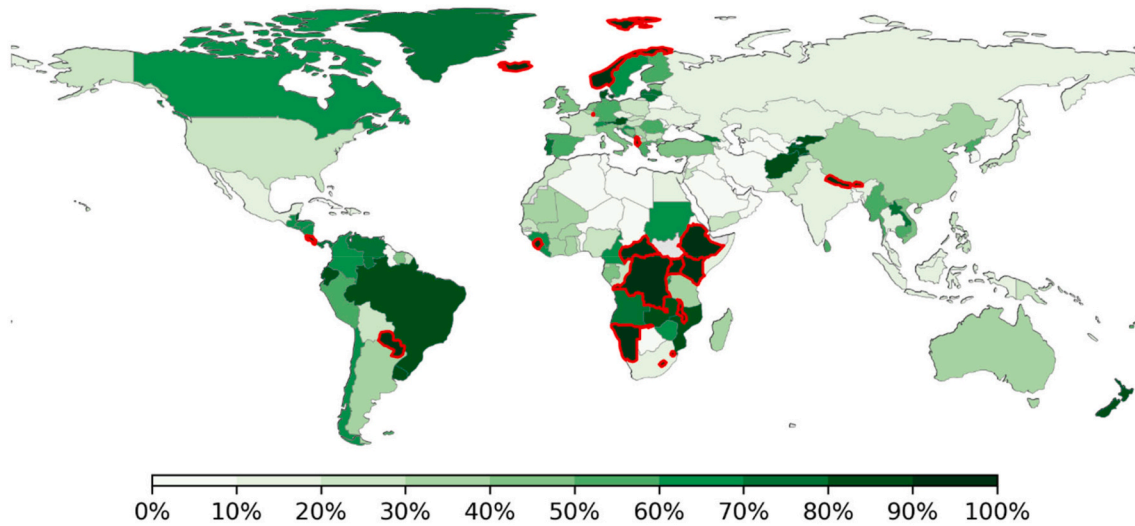


Fig. 1. Share of domestic electricity generation from low-carbon sources (renewables and nuclear), see Our World in Data [36] based on Ember [37]). Countries outlined in red have a low-carbon electricity share of 90% or more. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

from data centres, cryptocurrency mining, and rising air conditioning use further contribute to this trend. In this study, we prioritize domestic electricity demand over on-grid electrolysis for hydrogen and PtX export, recognizing the need to first meet domestic energy needs while efficiently using electricity directly [1]. The domestic electricity demand data is obtained from Fedotova et al. [38], which includes country-wise testing and validation of the power system modelling as presented in Ref. [40]. It is based on the demand time series generated using the *Synde* package [41], which was originally modelled by GlobalEnergyGIS following the approach by Mattsson et al. [42]. The domestic electricity demand of Fedotova et al. [38] is subsequently validated, scaled and extended with a demand trend aligned with the Stated Policy scenario of the International Energy Agency [43].

Notably, the STEPS scenario's electricity demand projections account for electrolytic hydrogen production within the industrial demand category, rather than excluding it as a separate exogenous load [43]. The domestic electricity demand applied in this study therefore already incorporates the (policy-consistent, comparatively modest) electricity requirements for domestic hydrogen and PtX production under stated policies, before this demand is subtracted from total electricity supply to determine the electricity surplus available for export-oriented hydrogen and PtX electrolysis (see “*Available electricity*” below). This approach avoids double-counting domestic hydrogen demand and ensures that only electricity exceeding both domestic electricity needs and domestically-assumed hydrogen production is allocated to export potential.

Renewable electricity expansion. The RE expansion is crucial to meet (growing) domestic demand and enable hydrogen exports, but deployment is constrained by workforce, supply chains, and grid infrastructure. Following Cherp et al. [44], we estimate maximum expansion rates $e_{t,c}$ for each technology t and country c using:

$$e_{t,c} = \frac{g_t}{100} \cdot d_{takeoff,t,c} \quad (1)$$

where g_t is the growth rate, and $d_{takeoff,t}$ is the demand when the technology surpasses 1% of total electricity supply. As proposed by Cherp et al. [44], we apply the upper expansion rates aligned with 1.5°C stabilization pathways: 1.4% per year for wind and 1.2% per year for solar. These rates represent historically observed upper bounds scaled to meet global climate targets and are applied uniformly across all countries as an optimistic yet physically grounded scenario. The demand at the take-off year $d_{takeoff,t}$ is determined country-wise from the dataset of

Fedotova et al. [38], while the take-off year is defined as the year in which a technology surpasses 1% of total electricity supply.

Applying these rates globally, the additional RE generation $e_{re,add,c}$ between years a_{target} and $a_{current}$ follows:

$$e_{re,add,c} = (a_{target} - a_{current}) \cdot \sum_t e_{t,c} \quad (2)$$

for each technology (solar, onshore wind) and country globally. While the expansion rates by Cherp et al. [44] do not consider land use constraints, we use Atlite [45] with weather and land availability data as showcased in Ref. [46] when validating the results using the energy model.

Threshold difference. To quantify the electricity available for hydrogen and PtX exports, we define the threshold difference (TD) representing the difference between the RE share s_{re} and the regulation threshold (T), scaled by total domestic demand e_{dom} :

$$TD = (s_{re} - T) \cdot e_{dom} \quad (3)$$

This means a country with a certain electricity demand and no RE expansion can export electricity equal to the difference between its current RE share and the hydrogen regulation threshold. Note that additional demands (as electricity for hydrogen exports) require the corresponding additional generation (RE or conventional) for the countries' energy balance to hold. To assess the impact of different thresholds on hydrogen exports, we apply a set of thresholds and investigate the impact of those in Section ‘The minimum share requirement dominates the export quantity’.

Available electricity. Based on the RE expansion and threshold difference, we define the available electricity for hydrogen exports $e_{exp,c}$ per country c as:

$$e_{exp,c} = e_{re,add,c} + TD - e_{dom,add,c} \quad (4)$$

subtracting the additional domestic electricity demand $e_{dom,add,c}$ (including domestic electricity demand for domestic hydrogen and PtX use) per country. This approach underlies the conservative assumption, that the future electricity demands are fully covered by RE. If this would not be the case, the increasing domestic electricity demand is not fully deducted from the available electricity for hydrogen exports, hence increasing the hydrogen export quantity.

The electrolysis efficiency $\eta_{electrolysis}$ is taken from the Danish Energy Agency (DEA) database [47] and varies by year, reflecting projected efficiency improvements (from 65% in 2025 to 72% by 2050, based on

lower heating value). Electrolyser deployment is not explicitly capacity-constrained in the global estimate, the analysis determines an upper-bound export potential, assuming that electrolyser capacity can be scaled to match available electricity for hydrogen export. This assumption is validated using the PyPSA-Earth model, which performs cost-minimizing capacity expansion including electrolyser sizing. Based on the electrolysis efficiency $\eta_{\text{electrolysis}}$, the subsequent hydrogen export quantity $h_{\text{exp},c}$ results in:

$$h_{\text{exp},c} = e_{\text{exp},c} \times \eta_{\text{electrolysis}} \quad (5)$$

2.2. Energy model

We validate the global, country-wise hydrogen and PtX quantities by deploying the energy model PyPSA-Earth to Ethiopia and further countries with high RE shares and a potential candidate for hydrogen and PtX exports under minimum RE share requirements. PyPSA-Earth is a global energy system model that provides detailed representations of the electricity, heat, gas, and transport sectors [46,48]. Built upon the open-source Python package PyPSA [49], it is designed to simulate energy systems with high spatial and temporal resolution. In this study, we apply the model to assess the effect of hydrogen and PtX exports under minimum RE share requirements in Ethiopia and further highly renewable countries. The model captures current and future energy systems, including fossil and nuclear technologies, and performs cost-minimizing optimization to determine the most efficient expansion and dispatch pathways under the given constraints.

3. Results

Based on the METHODS, this Section presents the main results. First, the impact of the minimum share requirement on the export quantity is described, analyzing the export potential in the near-term (2025). Second, this study investigates the near-, mid- and long-term potentials and shows how the Global export potentials peak in the mid-term. Lastly, the results are validated by a detailed modelling of countries with a high RE share.

3.1. The minimum share requirement dominates the export quantity

Fig. 2 shows the impact of minimum share requirements on export

quantities aggregated by continent, displaying the export quantity in the near-term (year 2025). If a 100% rule is applied, countries can only export hydrogen if the RE expansion in the year 2025 exceeds the domestic demand and electricity gap to achieve 100% (phasing out conventional power generation completely), which is only applicable for a few countries and continents worldwide and cumulating to 2.5 TWh in 2025. However, this changes with relaxing the minimum share requirement to 90%, enabling countries which are above 90% RE as of today (Fig. 1). Many of those countries are in Europe (incl. Norway) and South America, followed by countries on the African continent with high shares of RE (but smaller electricity systems and hence less export potential). The combined export potential under a 90% rule is 24 TWh globally (Fig. 2).

A further relaxation to 80% brings South American countries into scope, with Brazil (48 TWh) leading the way. Due to their high share of RE and large electricity system, a 80% rule would allow for substantial hydrogen exports up to 95 TWh globally. At a 60% rule, countries in North America (specifically Canada) come into play and are eligible for exporting up to 37 TWh. With this regulation, the hydrogen and PtX export quantity increases across all continents, dominated by South America (163 TWh), Europe (79 TWh) and North America (37 TWh). Also, Africa (20 TWh), Asia (14 TWh) and Oceania (8 TWh) can export substantial hydrogen and PtX quantities under a minimum renewable share regulation.

To conclude, the minimum RE share requirement dominates the export quantity. In the near-term (2025), a 90% rule unlocks global potentials of 24 TWh mostly in Europe and the South American continent, while a further relaxation of the regulation brings other world regions into play as well. A 70% rule would foster a combined global hydrogen and PtX quantity of up to 181 TWh.

3.2. Global export potentials peak in the mid-term

The previous section elaborates on the impact of the minimum share requirement on the export quantity in the near-term (2025). Based on those insights, this study investigates the short-, mid- and long-term (2050) hydrogen and PtX export potentials.

Fig. 3 shows the country-wise global export quantity from 2025 until 2050 under the 90% minimum RE share requirement. The violin plots indicate the median (white line), interquartile range (thick black bar), and 1.5x interquartile range (thin black bar) of the export potential per

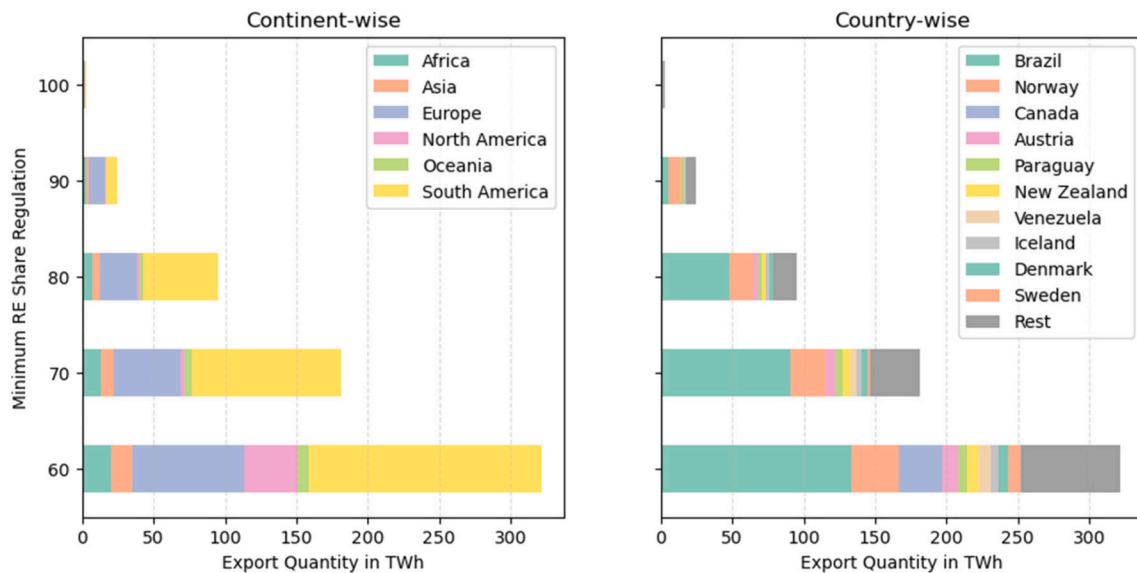


Fig. 2. The impact of minimum Renewable Electricity (RE) share requirements on export quantities aggregated by continent. Export quantities are in the near-term (year 2025), clustered by continent (right) and by country (left), including the 10 countries with the highest export quantities.

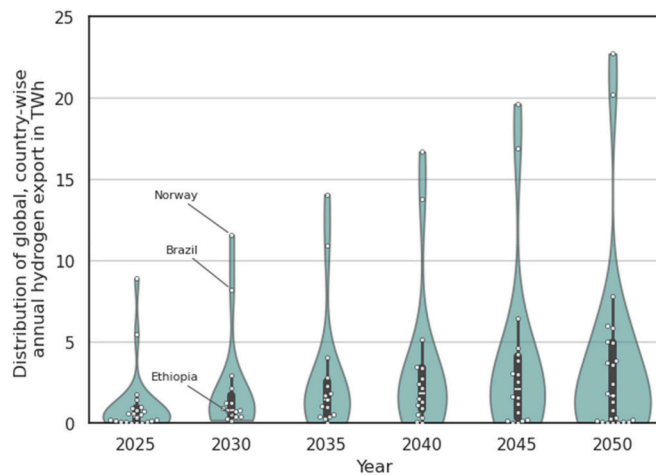


Fig. 3. Country-wise, global export quantity (90% rule). Violin plots indicate median (white line), interquartile range (thick black bar), and 1.5x interquartile range (thin black bar), using kernel density estimation to show the country-wise distribution of hydrogen and PtX export quantities. In addition, each country is represented as single white dot, showcased for Norway, Brazil and Ethiopia. Countries without hydrogen export quantities in the respective years are excluded from the dataset visualized.

country. Note that countries without hydrogen and PtX export quantities in the respective years are excluded from the dataset visualized, avoiding distortions in the distribution.

In the modelling year 2025, the annual hydrogen and PtX export potential median is 0.4 TWh, with a single country (Norway) peaking at 9 TWh (Fig. 3). The upper limit of the 1.5x interquartile range increases in the mid-term to 3 TWh (2030) and 4 TWh (2035), while the median stays below 2 TWh in those years. This means, that only a few countries can export substantial amounts of hydrogen and PtX (with peaks up to 14 TWh in 2035). This observation is rooted in the trend that the RE expansion outpaces the increase of domestic electricity demands, hence increasing the available electricity for hydrogen and PtX exports under the 90% regulation. However, half of the countries which have an export potential greater than 0 TWh, do not exceed an export potential of 2 TWh.

Five years later (2040), the characteristics are similar with only the maximum increasing from 14 TWh to 16 TWh (Fig. 3). In the long-term (2045 and 2050), the upper limit of 1.5x interquartile range reaches 7 TWh (2045) and 8 TWh (2050), respectively. Here, RE expansion rates outpace domestic electricity demands.

However, while the maximum export quantity per country and the upper limit of the 1.5x interquartile range increase, the median stays relatively constant at around 2 TWh. This means that more than 50% of the countries (which have an export potential above 0 TWh) can export 2 TWh or more. The global, total export quantity identified in Figs. 2 and 3 is put into context of global hydrogen demand scenarios in Section DISCUSSION.

To conclude, the findings highlight a shift in hydrogen and PtX export dynamics over time. While mid-term exports are driven by rapid renewable expansion outpacing domestic demand, the long-term trend reflects a growing number of exporting above a threshold of 1 TWh. This underscores the evolving balance between domestic electricity needs and hydrogen and PtX export potentials under the 90% minimum renewable share requirement.

3.3. Validation of export quantities in highly renewable electricity systems

To validate the feasibility of the global, country-wise export quantity, we deploy the energy model PyPSA-Earth (s. Sec. Energy model) to various countries with highly RE systems on different continents

worldwide.

Fig. 4 shows the cost-optimized electricity generation and transmission capacities in Ethiopia in 2030, with additional electricity demands of 30% (compared to the base scenario in 2030). This additional electricity demand is equivalent to a hydrogen or PtX export quantity of 3 TWh. In Ethiopia, the electricity system is dominated by hydro power plants, while the additional demand imposed on the system increases the onshore wind deployment in the centre of Ethiopia and minor additional solar PV installations in the North.

In addition to the detailed analysis of Ethiopia, this study examines hydrogen and PtX export potentials across a range of countries at high temporal (3-hourly) and spatial resolution, based on scenarios with a 30% exogenous increase in electricity demand. To facilitate access to country-level results, an open-source, publicly available dashboard has been developed: <https://exportquan.streamlit.app/>. This dashboard is continuously updated and expanded to support further exploration of the findings.

4. Discussion

Based on the results, the discussion contextualizes the relevance of a minimum RE share requirement on global hydrogen and PtX export quantities, with strong implications for policy design and energy transition strategies.

The impact of minimum share requirements on global hydrogen and PtX exports. The analysis demonstrates that minimum RE share requirements shape the global hydrogen and PtX export landscape. In the near-term (2025), a strict 90% or 100% requirement limits export potential to only a handful of countries, resulting in a mere 2.5 to 24 TWh of exportable hydrogen. However, relaxing the threshold to 80% unlocks substantial export quantities — up to 95 TWh — primarily driven by countries in South America and Europe. This regulatory lever, therefore, emerges as a possibility for countries with a current high share of RE and large electricity systems in the early years.

From a broader perspective, even under optimistic assumptions (e.g. 70% threshold), the maximum global export potential in 2025 is 181 TWh. This quantity is relatively modest compared to projected global hydrogen demands of 2000 TWh by 2030 projected by net-zero scenarios of the International Energy Agency [50]. While the minimum share regulation can help mobilize early supply, it may only provide a small fraction of total global needs.

Time scale: From near-term enabler to long-term sustainable growth. Furthermore, the results reveal the potential to unlock hydrogen and PtX exports in the short term. Compared to the other regulatory requirements such as additionality, temporal and geographical correlation, the simplicity of the minimum share requirements potentially encourages investments in electrolyzers and jumpstarts hydrogen and PtX trade in renewable-rich countries. While blue hydrogen pathways might enable a quick scale up of hydrogen production via steam methane reforming and carbon capture, questions on the cost-competitiveness (taking CO₂-eq prices into account) and on the danger of fossil lock-ins remain [13]. Unlike blue hydrogen pathways, a minimum share requirement is compatible with long-term green hydrogen targets as it promotes the installation of hydrogen electrolysis. If complemented with policies on additionality, temporal and geographical matching in the long-term, this could ensure that hydrogen and PtX export pathways remain sustainable and genuinely low-carbon over time.

In the medium term (2030–2035), hydrogen and PtX export quantities are projected to increase, driven by renewable expansion outpacing domestic demand growth. This suggests that a minimum RE share regulation could open a window of opportunity where excess renewable generation is used for hydrogen production. However, the efficiency of hydrogen applications (especially when direct electrification options exist) should be considered to ensure effective use of renewable energy [1].

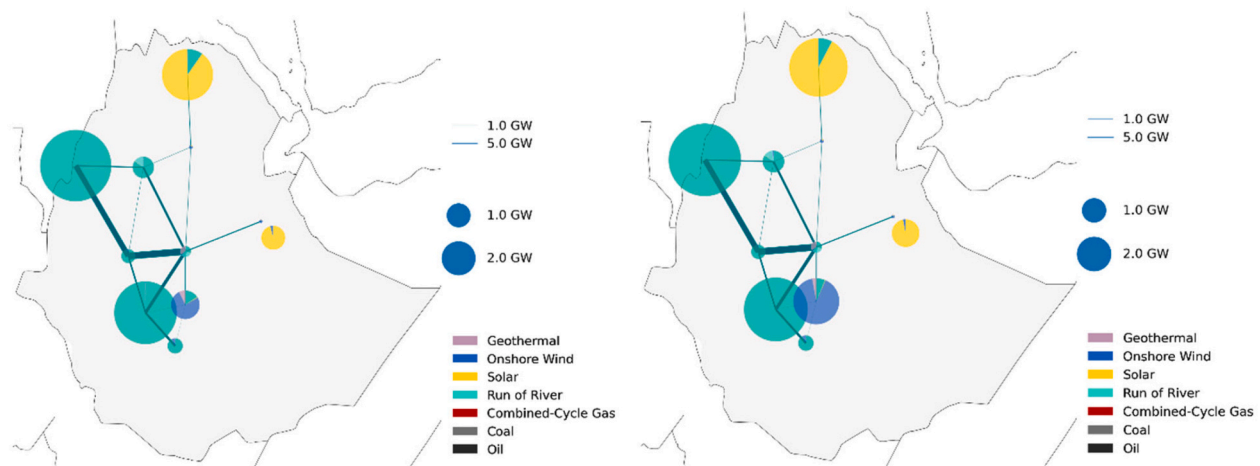


Fig. 4. Optimized electricity generation and transmission capacities in Ethiopia in 2030, without hydrogen exports (left) and with hydrogen exports equivalent to additional electricity demands of 30% (compared to the base scenario in 2030) validating the feasibility of electricity demands for hydrogen exports.

Differentiating between short-term and medium-term policy implications is critical for strategic planning. Countries that already exhibit exceptionally high renewable electricity shares (such as New Zealand or Ethiopia (>90%)) possess a distinct near-term advantage. In the short term (up to 2030), these regions can facilitate early and rapid investment decisions for electrolyser deployments. Because they already meet stringent renewable share thresholds, the initial capital expenditure is significantly lower compared to regions that must simultaneously finance and construct new, dedicated renewable capacities to satisfy additionality rules.

However, in the medium term (2030–2035), policy priorities shift. To sustain green hydrogen and PtX production alongside growing domestic electrification, continuous and substantial renewable deployment strategies remain necessary. Failing to maintain high expansion rates jeopardizes the long-term compliance of domestic decarbonization efforts and hydrogen exports.

Threshold trade-offs: balancing ambition, feasibility, and emissions. The minimum share regulation, while effective at steering supply toward cleaner sources, also raises concerns about consequential greenhouse gas emissions. Stricter thresholds like 90% to 100% guarantee low emission intensity but restrict eligibility to only a few countries, increase hydrogen and PtX production cost and thus lower the economic breakthrough. In contrast, more relaxed thresholds like 70% broaden participation but raise concerns about the actual carbon footprint of hydrogen and PtX exports, particularly in systems where fossil generation still plays a role.

To illustrate the trade-off between high minimum RE share requirements of 90% versus 70% quantitatively, we determine the specific hydrogen emissions for both cases in Table 1. We consider an electrolyser with 62% efficiency, requiring 53.8 kWh of electricity per kg H₂, based on hydrogen's lower heating value of 33.33 kWh/kg. Assuming the residual fossil share is gas-fired (490 gCO₂/kWh, based on the IPCC AR5 lifecycle median [51]), the resulting specific emissions of hydrogen

are 2.6 kg CO₂/kgH₂ at a 90% renewable-share threshold (Case A), compared with approximately 7.9 kg CO₂/kgH₂ at the more permissive 70% threshold (Case B), see Table 1. While both values undercut the carbon intensity of grey hydrogen (10 kg CO₂/kg H₂) by 74% and 21%, respectively, only the stricter 90% threshold yields a carbon intensity below the 4 kg CO₂/kgH₂ cut-off applied in the U.S. 45V tax credit scheme. If the non-renewable were coal-fired (emission intensity of 820 gCO₂/kWh [51]) rather than gas-fired, the resulting specific hydrogen emissions would increase by 67% each, given coal's substantially higher emission factor.

On the one hand, how the electricity demand for hydrogen is counterbalanced within the national electricity system (or bidding zone) is crucial. Even in systems that meet a 90% renewable share, additional hydrogen production may trigger the dispatch of fossil generation to maintain energy balance [25]. On the other hand, if electrolysers are placed near sites with high RE curtailment, the energy system benefits by integrating unused RE shares, lessening stress on the electricity grids and overall emissions and lowering costs for electricity, when electrolysers provide additional flexibility in distribution grids [52].

Complementing the analysis of specific emissions of hydrogen, total energy system emissions are only effectively mitigated if domestic decarbonization efforts are not in conflict with hydrogen and PtX export ambitions. While rapid renewable expansion can create temporary surplus enabling hydrogen exports, rising domestic electricity demand from electrification, economic growth and data centres (as discussed by Pabon et al. [53]) progressively reduces this surplus over time. This implies that sustained hydrogen export strategies depend not only on renewable resource availability, but also on the pace of domestic demand growth and the prioritisation of local decarbonization objectives as outlined in the section Results. Consequently, hydrogen and PtX export scenarios must be evaluated in the context of long-term domestic energy system transformation and potential competition for renewable electricity [26], as we do in this study.

Limitations and further research. In addition, further research is required, and certain limitations must be acknowledged. First, data gaps regarding electricity demand projections, renewable share trajectories, and country-level specifics introduce uncertainties, particularly in the long-term projections. Second, the assumption that each country acts as a single bidding zone without limitations in electricity grids overlooks sub-national disparities and lack of network expansion, especially in geographically large or federally structured countries. This may overestimate export potentials (in countries part of a larger bidding zone) or underestimate export potentials (in countries divided into further bidding zones such as Denmark, where the bidding zone DK1 is close to 90% RE share already). Third, while this study focuses on the supply side, the

Table 1

Quantitative assessment of specific hydrogen emissions based on electricity systems with 90% (Case A) or 70% (Case B) renewable electricity (RE) share. Both values have substantially lower carbon intensity than grey hydrogen (10 kg CO₂/kg H₂).

Case	RE share	Gas share	Grid emission factor	GHG emissions of H ₂
A	90%	10%	10% * 490 g/kWh = 49 g/kWh	2.6 kgCO ₂ /kgH ₂
B	70%	30%	30% * 490 g/kWh = 147 g/kWh	7.9 kgCO ₂ /kgH ₂

analysis omits hydrogen and PtX trade and demands for hydrogen imports. Lastly, this study does not account for electricity imports or exports and their potential impact on national renewable shares. Instead, we assume that the renewable share is calculated based on domestic generation alone.

It is important to note that this study assumes isolated, country-level electricity systems. In reality, grid constraints and cross-border interconnectors may influence the feasibility and cost-effectiveness of hydrogen production. Intra-national transmission bottlenecks can limit deployment of remote, high-yield renewable sites with electrolysis plants (if bottlenecks between electricity supply and demand exist) while cross-border trade can facilitate the balancing of intermittent renewable supply.

It must be emphasized that the export potentials calculated in this study represent theoretical upper bounds, constrained primarily by land availability, renewable capacity factors, and regulatory criteria. Realizing these volumes will heavily depend on overcoming substantial infrastructure bottlenecks across the supply chain. The temporal mismatch between intermittent renewable generation and continuous baseload export logistics necessitates cost-effective buffer options, such as large-scale hydrogen underground storage [54]. Furthermore, downstream transportation constraints will influence practical export volumes. These include the current lack of dedicated hydrogen transmission pipelines and costs associated with converting hydrogen into PtX products like ammonia or synthetic natural gas (SNG) for maritime shipping. Consequently, while this study establishes the foundation of generation eligibility, future spatial modelling is required to explicitly integrate these logistical components to translate theoretical capacities into practical export scenarios.

Beyond technical and regulatory constraints, the realization of the identified export potentials will be heavily shaped by geopolitical and economic dynamics. The transition from the theoretical capacity to active international trade requires substantial capital investments, which are contingent upon stable geopolitical environments, long-term off-take agreements, and de-risking mechanisms. Long-term market ramp-up will rely on the development of fair partnerships that not only serve the import demand of industrialized regions but also foster local economic development, domestic energy security, and industrial capacity in the exporting countries.

While PyPSA-Earth provides a comprehensive framework for modelling global hydrogen potentials, the robustness of the model outputs is subject to inherent uncertainties. Rather than performing a quantitative uncertainty analysis across all global nodes, it is crucial to acknowledge the key sensitivities driving these results. The most prominent uncertainties stem from future cost assumptions (e.g., capital expenditures for electrolyzers and renewable generation technologies) and varying projections for domestic energy demands. Changes in these underlying assumptions could significantly influence the calculated export potentials and must be considered when interpreting the results for policy-making. Therefore, we have provided all our code, data and input assumptions as open source/data, as specified in the data availability statement.

Further research could expand on these findings by quantifying consequential emissions and their underlying drivers, mechanisms, and policy frameworks to avoid carbon leakage. In addition, a further validation for a broader set of countries and global renewable energy potential limits would strengthen the robustness of the results. Sensitivity analyses on electricity demand assumptions and the inclusion of trade and RE shares accounting could offer deeper insights into export eligibility and system interactions.

5. Conclusion

This study shows how minimum RE share requirements shape the global landscape for hydrogen and PtX exports. Stricter thresholds, such as 90%, limit near-term export eligibility to a few countries, while more

relaxed rules significantly expand global potential. Export volumes are projected to increase in the medium term (2030–2035), when renewable expansion temporarily outpaces electricity demand growth, highlighting a time-sensitive window of opportunity. The minimum share regulation unlocks short-term hydrogen and PtX export potential while avoiding fossil lock-ins with blue or grey hydrogen. While the additional electricity demand for hydrogen electrolysis may be supplied by RE, the risk that the additional demand is indirectly met by ramping up fossil-based generation, even in high-RE systems, underscores the need for robust verification and legislation to ensure low consequential emissions. Our country-level analysis fills a gap in the literature by quantifying the global hydrogen and PtX production and export potential under minimum share requirements (an aspect often overlooked in previous studies) and demonstrates it in a case study of Ethiopia and further countries worldwide.

Looking ahead, hydrogen regulation will be essential to scale hydrogen and PtX supply while safeguarding climate integrity. By aligning export growth with ambitious RE targets, minimum share requirements can serve as a pragmatic lever, helping hydrogen-exporting countries realize their potential and create value, accelerating the global transition to net-zero emissions.

Data and code availability

The data and code used in this study are accessible and reproducible at <https://github.com/energyLS/exportquan> and archived as version v0.0.1 on Zenodo accessible at <https://doi.org/10.5281/zenodo.20644263>. The code used in this study is reproducible as instructed on <https://github.com/energyLS/exportquan/blob/main/README.md>.

Furthermore, a publicly available dashboard has been developed in the same codebase and is visible at: <https://exportquan.streamlit.app/>.

CRedit authorship contribution statement

Leon Schumm: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Anton Achhammer:** Data curation, Formal analysis, Software, Validation, Writing – original draft. **Michael Sterner:** Funding acquisition, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors are grateful for funding from the H2Global meets Africa project (03SF0703A) by the German Federal Ministry of Research, Technology and Space and for helpful comments by Marco Schamel.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used DeepL and ChatGPT in order to improve wording. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

References

- [1] Ueckerdt F, Bauer C, Dirnhaichner A, Everall J, Sacchi R, Luderer G. Potential and risks of hydrogen-based e-fuels in climate change mitigation. *Nat Clim Change* 2021;11(5):384–93. <https://doi.org/10.1038/s41558-021-01032-7>.
- [2] Gailani A, Cooper S, Allen S, Pimm A, Taylor P, Gross R. Assessing the potential of decarbonization options for industrial sectors. *Joule* Jan. 2024;0(0). <https://doi.org/10.1016/j.joule.2024.01.007>.
- [3] Kovač A, Paranos M, Marcius D. Hydrogen in energy transition: a review. *Int J Hydrogen Energy* Mar. 2021;46(16):10016–35. <https://doi.org/10.1016/j.ijhydene.2020.11.256>.
- [4] Azadnia AH, McDaid C, Andwari AM, Hosseini SE. Green hydrogen supply chain risk analysis: a european hard-to-abate sectors perspective. *Renew Sustain Energy Rev* Aug. 2023;182:113371. <https://doi.org/10.1016/j.rser.2023.113371>.
- [5] Sterner M, Specht M. Power-to-Gas and Power-to-X—The history and results of developing a new storage concept. *Energies* Jan. 2021;14(20). <https://doi.org/10.3390/en14206594>.
- [6] International Energy Agency. Net zero by 2050. A roadmap for the Global energy sector. INTERNATIONAL ENERGY AGENCY; 2021.
- [7] Sterner M, et al. 19 Import options for green hydrogen and derivatives - an overview of efficiencies and technology readiness levels. *Int J Hydrogen Energy* Nov. 2024;90:1112–27. <https://doi.org/10.1016/j.ijhydene.2024.10.045>.
- [8] Lo Basso G, et al. Status and perspectives of hydrogen role for decarbonising industry: a comprehensive review. *Renew Sustain Energy Rev* Dec. 2025;224:116083. <https://doi.org/10.1016/j.rser.2025.116083>.
- [9] Genovesi M, Corigliano O, Piraino F, Fragiaco P. E-fuels infrastructures for heavy-duty applications: case study of a refueling facility based on green hydrogen. *Int J Hydrogen Energy* Jul. 2025;144:1367–82. <https://doi.org/10.1016/j.ijhydene.2025.03.423>.
- [10] Cvetkovska R, Wechner L, Kienberger T. Techno-economic assessment of hydrogen supply solutions for industrial site. *Int J Hydrogen Energy* Feb. 2025;104:611–22. <https://doi.org/10.1016/j.ijhydene.2024.09.099>.
- [11] Odenweller A, Ueckerdt F. The green hydrogen ambition and implementation gap. *Nat Energy* Jan. 2025;10(1):110–23. <https://doi.org/10.1038/s41560-024-01684-7>.
- [12] Wu W, Zhai H, Holubnyak E. Technological evolution of large-scale blue hydrogen production toward the U.S. hydrogen energy earthshot. *Nat Commun* Jul. 2024;15(1):5684. <https://doi.org/10.1038/s41467-024-50090-w>.
- [13] Ueckerdt F, et al. On the cost competitiveness of blue and green hydrogen. *Joule* Jan. 2024;8(1):104–28. <https://doi.org/10.1016/j.joule.2023.12.004>.
- [14] Neumann F, Hamp J, Brown T. Energy imports and infrastructure in a carbon-neutral European energy system. *arXiv: arXiv:2404.03927* Apr. 05, 2024. <https://doi.org/10.48550/arXiv.2404.03927>.
- [15] Commission Delegated Regulation (EU) 2023/1184 of 10 February 2023 supplementing Directive (EU) 2018/2001 of the European Parliament and of the Council by establishing a Union methodology setting out detailed rules for the production of renewable liquid and gaseous transport fuels of non-biological origin, vol. 157. 2023. Accessed: Dec. 27, 2024. [Online]. Available: <http://data.europa.eu/eli/reg/del/2023/1184/oj/eng>.
- [16] L. Collins, ‘Germany wants EU to postpone controversial additionality rules “so green hydrogen can be produced more cheaply”’, *hydrogeninsight.com*. Accessed: December. 19, 2024. [Online]. Available: <https://www.hydrogeninsight.com/policy/germany-wants-eu-to-postpone-controversial-additionality-rules-so-green-hydrogen-can-be-produced-more-cheaply-2-1-1712665>.
- [17] Copenhagen infrastructure partners. Grüner wird’s nicht 2024 [Online]. Available: https://www.ost-ausschuss.de/sites/default/files/eventdocs/2024_12_Gr%C3%BCner%20wird%27s%20nicht.pdf.
- [18] de Jong C, Kazmi E. Green hydrogen business cases. *Kyos* Oct 2023;31 [Online]. Available: <https://www.kyos.com/wp-content/uploads/2023/11/KYOS-hydrogen-webinar-20231031.pdf>.
- [19] Mazzeo D, Herdem MS, Matera N, Wen JZ. Green hydrogen production: analysis for different single or combined large-scale photovoltaic and wind renewable systems. *Renew Energy* Nov. 2022;200:360–78. <https://doi.org/10.1016/j.renene.2022.09.057>.
- [20] Bhandari R. Green hydrogen production potential in West Africa – case of Niger. *Renew Energy* Aug. 2022;196:800–11. <https://doi.org/10.1016/j.renene.2022.07.052>.
- [21] Colbertaldo P, Guandalini G, Campanari S. Modelling the integrated power and transport energy system: the role of power-to-gas and hydrogen in long-term scenarios for Italy. *Energy* Jul. 2018;154:592–601. <https://doi.org/10.1016/j.energy.2018.04.089>.
- [22] Liponi A, Baccioli A, Ferrari L. Feasibility analysis of green hydrogen production from wind. *Int J Hydrogen Energy* Dec. 2023;48(96):37579–93. <https://doi.org/10.1016/j.ijhydene.2023.05.054>.
- [23] Henry A, McStay D, Rooney D, Robertson P, Foley A. Techno-economic analysis to identify the optimal conditions for green hydrogen production. *Energy Convers Manag* Sep. 2023;291:117230. <https://doi.org/10.1016/j.enconman.2023.117230>.
- [24] Iileka S, Likius DS, Tomas P, Chembe DK, MuAshekele H. Viability of green hydrogen production at Benguela wind energy community Project in Lüderitz, Namibia. [Journal of Sustainable Development of Energy, Water and Environment Systems] Sep. 2024;12(3):1–13 [Online]. Available: <https://www.sdwes.org/jsdewes/paper.php>. [Accessed 8 April 2025].
- [25] Schumm L, Haas J, Peer R, Sterner M. The role of hydrogen offtaker regulation in highly renewable electricity systems. *Energy* Jan. 2026;342:139513. <https://doi.org/10.1016/j.energy.2025.139513>.
- [26] Schumm L, et al. The impact of temporal hydrogen regulation on hydrogen exporters and their domestic energy transition. *Nat Commun* Aug. 2025;16(1):7486. <https://doi.org/10.1038/s41467-025-62873-w>.
- [27] Müller VP, Eichhammer W, van Vuuren D. Paving the way: analysing energy transition pathways and green hydrogen exports in developing countries – the case of Algeria. *Int J Hydrogen Energy* May 2024;67:240–50. <https://doi.org/10.1016/j.ijhydene.2024.04.153>.
- [28] Zeyen E, Riepin I, Brown T. Temporal regulation of renewable supply for electrolytic hydrogen. *Environ Res Lett* Feb. 2024;19(2):024034. <https://doi.org/10.1088/1748-9326/ad2239>.
- [29] Ruhnau O, Schiele J. Flexible green hydrogen: the effect of relaxing simultaneity requirements on project design, economics, and power sector emissions. *Energy Policy* Nov. 2023;182:113763. <https://doi.org/10.1016/j.enpol.2023.113763>.
- [30] Brauer J, Villavicencio M, Trüby J. Green hydrogen – how grey can it be? European University Institute Robert Schuman Centre for Advanced Studies The Florence School of Regulation; 2022 [Online]. Available: https://cadmus.eui.eu/bitstream/handle/1814/74850/RSC_WP_2022_44.pdf.
- [31] Giovanniello MA, Cybulsky AN, Schittekatte T, Mallapragada DS. The influence of additionality and time-matching requirements on the emissions from grid-connected hydrogen production. *Nat Energy* Jan. 2024;1–11. <https://doi.org/10.1038/s41560-023-01435-0>.
- [32] Linsel Oliver, Bertsch Valentin. Effects of the delegated act on renewable fuels of non-biological origin on production costs and distribution of hydrogen production capacities across Europe. *Energy Policy* 2026;212:115178. <https://doi.org/10.1016/j.enpol.2026.115178>. <https://linkinghub.elsevier.com/retrieve/pii/S0301421526001126>.
- [33] Brandt J, et al. Cost and competitiveness of green hydrogen and the effects of the European Union regulatory framework. *Nat Energy* Jun. 2024;9(6):703–13. <https://doi.org/10.1038/s41560-024-01511-z>.
- [34] Delft CE. Additionality of renewable electricity for green hydrogen production in the EU. 2022 [Online]. Available: https://cedelft.eu/wp-content/uploads/sites/2/2022/12/CE_Delft_220358_Final_report.pdf. [Accessed 25 July 2024].
- [35] Hydrogen Europe. Summary and implication of the proposed Delegated Act on additionality of renewable for the production of RFNBs under the Renewable Energy Directive. 2022 [Online]. Available: <https://hydrogeneurope.eu/wp-content/uploads/2022/05/220523-Summary-DA-Additionality-final.pdf>. [Accessed 11 June 2026].
- [36] Our World in Data, ‘Share of electricity generated by low-carbon sources’, *Our World in Data*. Accessed: April. 9, 2025. [Online]. Available: <https://ourworldindata.org/grapher/share-electricity-low-carbon>.
- [37] EMBER. Global Electricity Review 2024 [Online]. Available: <https://ember-energy.org/app/uploads/2024/05/Report-Global-Electricity-Review-2024.pdf>; 2024.
- [38] Fedotova E. A global dataset of scaling coefficients for 2050 electricity demand projections. 2025. <https://doi.org/10.5281/zenodo.15178540>.
- [39] ISO 3166-1 alpha-2. Apr. 04. Wikipedia. 2025 [Online]. Available: https://en.wikipedia.org/w/index.php?title=ISO_3166-1_alpha-2&oldid=1283899224. [Accessed 7 April 2025].
- [40] Fioriti D, et al. Country-Wise open energy planning in high-resolution with PyPSA-Earth. In: 2023 international conference on smart energy systems and technologies (SEST); Sep. 2023. p. 1–6. <https://doi.org/10.1109/SEST57387.2023.10257559>.
- [41] Hamp J. Euronion/synde. Jupyter notebook. Oct. 29, 2024. Accessed: Apr. 09, 2025. [Online]. Available: <https://github.com/euronion/synde>.
- [42] Mattsson N, Verendel V, Hedenus F, Reichenberg L. An autopilot for energy models – automatic generation of renewable supply curves, hourly capacity factors and hourly synthetic electricity demand for arbitrary world regions. *Energy Strategy Rev* Jan. 2021;33:100606. <https://doi.org/10.1016/j.esr.2020.100606>.
- [43] IEA. Global energy and climate model documentation 2024. 2024 [Online]. Available: <https://iea.blob.core.windows.net/assets/89a1aa9a-e1bd-4803-b37b-59d6e7fba1e9/GlobalEnergyandClimateModelDocumentation2024.pdf>.
- [44] Cherp A, Vinichenko V, Tosun J, Gordon JA, Jewell J. National growth dynamics of wind and solar power compared to the growth required for global climate targets. *Nat Energy* Jul. 2021;6(7):742–54. <https://doi.org/10.1038/s41560-021-00863-0>.
- [45] Hofmann F, Hamp J, Neumann F, Brown T, Hörsch J. Atlite: a lightweight python package for calculating renewable power potentials and time series. *J Open Source Softw* Jun. 2021;6(62):3294. <https://doi.org/10.21105/joss.03294>.
- [46] Parzen M, et al. PyPSA-Earth. A new global open energy system optimization model demonstrated in Africa. *Appl Energy* Jul. 2023;341:121096. <https://doi.org/10.1016/j.apenergy.2023.121096>.
- [47] PyPSA Community, Danish Energy Agency. PyPSA/technology-data: Technology Data v0.6.2. Energy Reports 2023. <https://doi.org/10.5281/zenodo.8220602>. <https://zenodo.org/records/8220602>.
- [48] Abdel-Khalek H, Schumm L, Jalbout E, Parzen M, Schau C, Fioriti D. PyPSA-Earth sector-coupled: a global open-source multi-energy system model showcased for hydrogen applications in countries of the Global South. *Appl Energy* Apr. 2025;383:125316. <https://doi.org/10.1016/j.apenergy.2025.125316>.
- [49] Brown T, Hörsch J, Schlachtberger D. PyPSA: python for power system analysis. *J Open Res Software* 2018;6. <https://doi.org/10.5334/jors.188>.
- [50] Net Zero roadmap: a global pathway to keep the 1.5 °C goal in reach – analysis’, IEA. Accessed: April. 8, 2025. [Online]. Available: <https://www.iea.org/reports/net-zero-roadmap-a-global-pathway-to-keep-the-15-oc-goal-in-reach>.
- [51] IPCC Working Group III. Annex III technology-specific cost and performance parameters. 2014.
- [52] Luderer G, Brown T, Bartels F. Report: die Energiewende kosteneffizient gestalten – szenarien zur Klimaneutralität 2045 | Ariadne [Online]. Available: <https://aria>

- dneprojekt.de/publikation/report-szenarien-zur-klimaneutralitat-2045/. [Accessed 9 April 2025].
- [53] Pabon JJG, Wang J, Chamanepour E, Salami D, Khosravi A. Dynamic integration of solar-powered hydrogen systems with fuel cells and district heating for green data centers. *Int J Hydrogen Energy* Dec. 2025;196:152557. <https://doi.org/10.1016/j.ijhydene.2025.152557>.
- [54] Achhammer A, Fioriti D, Patonia A, Sterner M. The impact of hydrogen underground storage on fair partnerships: a GIS-based integration of salt caverns into PyPSA-Earth. Feb. 26 Soc Sci Res Netw 2026:6307406. <https://doi.org/10.2139/ssrn.6307406>. Rochester, NY.